STATISTICS & MACHINE LEARNING · LESSON 16 OF 28

K-Nearest Neighbours: Classifier and Regressor (Bank Loans Case Study)

The simplest learning algorithm: predict from the most similar past examples. Distance metrics, choosing k, why scaling is essential, weighted voting, the curse of dimensionality, KNN for classification and regression, tuning with GridSearchCV, and the full Bank Loans case study from the live class with metrics and scoring of new applicants.

ML class 7 · scikit-learn 1.7+ · Intermediate · 100 min

By the end of this lesson you can

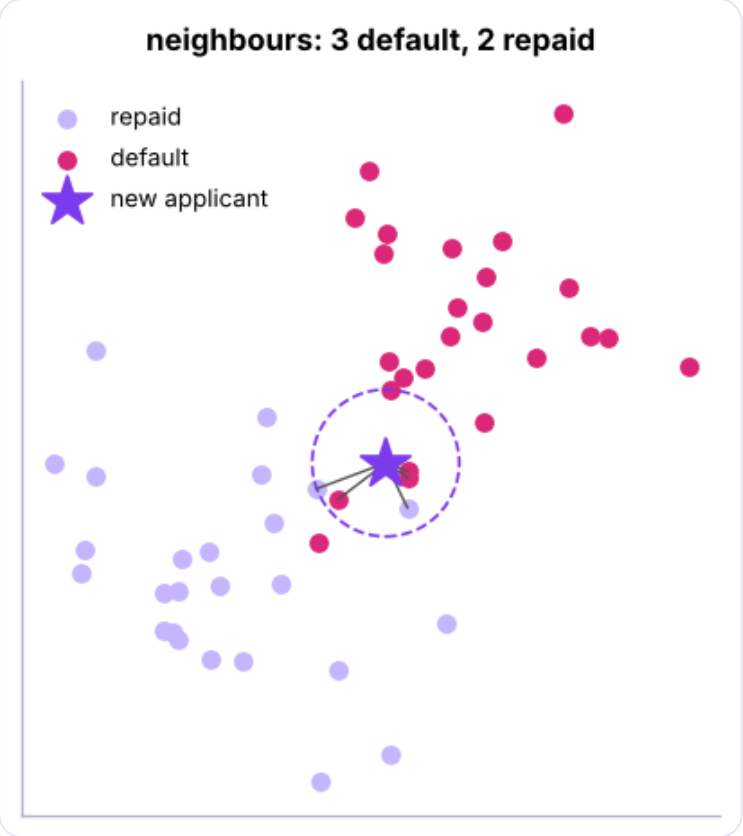
- Explain how KNN classifies and regresses using distances to neighbours
- Compute Euclidean, Manhattan and Minkowski distances and know why scaling matters
- Choose k and weighting with cross-validation and explain the bias-variance effect of k
- Run the Bank Loans case study end to end with a pipeline and GridSearchCV
- Describe KNN's strengths, weaknesses and the curse of dimensionality

Datasets used

`bank_loans.csv` `house_prices.csv`

The idea

"Tell me who your neighbours are, and I'll tell you who you are." To predict a new point, find the *k* most similar labelled points and let them **vote** (classification) or **average** their values (regression). There is no training beyond storing the data, so KNN is a **lazy, instance-based** learner.



KNN with k = 5: the new applicant (star) is classified by the majority of its five nearest neighbours.

Distances

$$d_{\text{Euclidean}}(a,b) = \sqrt{\sum_j (a_j - b_j)^2} \qquad d_{\text{Manhattan}}(a,b) = \sum_j |a_j - b_j|$$

Minkowski distance generalises both (`p=2` Euclidean, `p=1` Manhattan). Cosine distance suits text and embeddings; Hamming suits binary features.

Python

```python
from scipy.spatial import distance
x1, x2 = np.array([30, 45.0, 8.5]), np.array([42, 60.0, 12.0])          # age, income (thousands), debt-to-income %
print("Euclidean", round(distance.euclidean(x1, x2), 2), "| Manhattan",
distance.cityblock(x1, x2), "| cosine", round(distance.cosine(x1, x2), 4))
```

OUTPUT

```
Euclidean 19.53 | Manhattan 30.5 | cosine 0.0003
```

Why scaling is essential

Python

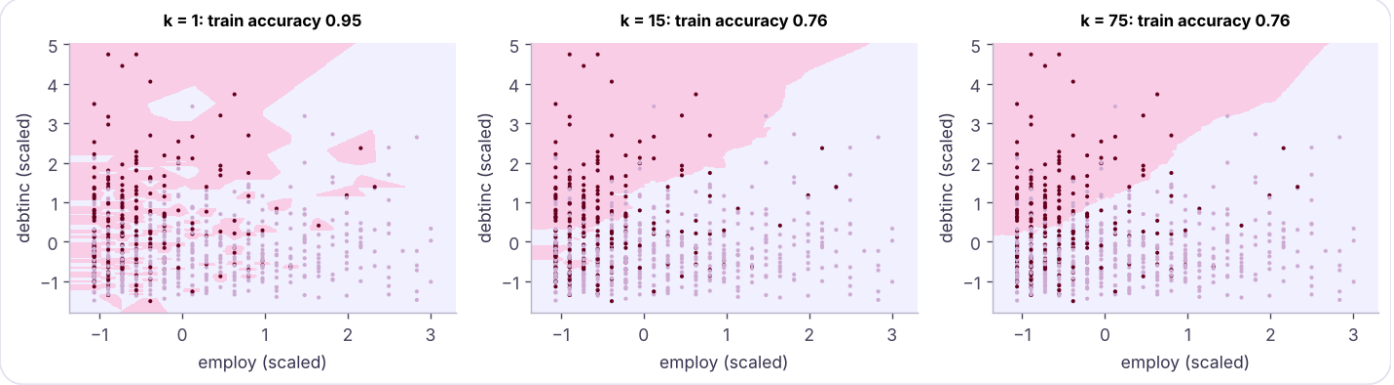
```
b = pd.read_csv("bank_loans.csv")
known = b[b.default.notna()].copy(); known["default"] = known.default.astype(int); new_apps
= b[b.default.isna()].copy()
feats = ["age", "ed", "employ", "address", "income", "debtinc", "creddebt", "othdebt"]
print(known[feats].describe().loc[["mean", "std", "max"]].round(1))
```

OUTPUT

	age	ed	employ	address	income	debtinc	creddebt	othdebt
mean	37.4	1.9	6.3	5.3	49.5	10.0	1.7	3.3
std	10.6	1.0	5.9	5.0	24.9	6.5	1.7	3.4
max	55.0	5.0	24.0	22.0	172.4	41.0	12.3	43.0

Income ranges up to 170 while `ed` ranges 1–5: unscaled, income dominates every distance and education is ignored. Standardise (or min-max scale) before KNN.

Choosing k



Decision boundaries for k = 1, 15 and 75 on two standardised features: small k overfits with jagged islands; large k oversmooths.

Small k	Large k
Flexible, low bias, high variance (overfits)	Smooth, high bias, low variance (underfits)
k = 1 memorises training data (100% train accuracy)	k = n predicts the majority class everywhere

Rules of thumb: start near √n, prefer odd k for binary problems (no ties), and choose by cross-validation. `weights="distance"` lets closer neighbours count more.

Case study: Bank Loans

Business problem: decide which of 150 new loan applicants to approve, using 700 past customers with known outcomes.

Python

```
from sklearn.model_selection import train_test_split, GridSearchCV, StratifiedKFold
from sklearn.metrics import roc_auc_score, classification_report, confusion_matrix,
accuracy_score
X, y = known[feats], known.default
Xtr, Xte, ytr, yte = train_test_split(X, y, test_size=0.3, stratify=y, random_state=42)
raw = KNeighborsClassifier(15).fit(Xtr, ytr); scaled = make_pipeline(StandardScaler(),
KNeighborsClassifier(15)).fit(Xtr, ytr)
print(f"k=15 without scaling: AUC {roc_auc_score(yte, raw.predict_proba(Xte)[:, 1]):.3f} |
with scaling: AUC {roc_auc_score(yte, scaled.predict_proba(Xte)[:, 1]):.3f}")
```

OUTPUT

```
k=15 without scaling: AUC 0.757 | with scaling: AUC 0.765
```

Scaling helps only a little here, because the large-range features (income, debt-to-income) happen to be informative anyway. When a large-range feature is irrelevant (for example a customer ID number or a postcode), unscaled KNN can fail completely.

Python

```
pipe = make_pipeline(StandardScaler(), KNeighborsClassifier())
grid = {"kneighborsclassifier__n_neighbors": list(range(3, 62, 4)),
"kneighborsclassifier__weights": ["uniform", "distance"], "kneighborsclassifier__metric":
["euclidean", "manhattan"]}
gs = GridSearchCV(pipe, grid, cv=StratifiedKFold(5, shuffle=True, random_state=0),
scoring="roc_auc", n_jobs=-1).fit(Xtr, ytr)
print("best:", {k.split("__")[1]: v for k, v in gs.best_params_.items()}, f"| CV AUC
{gs.best_score_:.3f}")
p = gs.predict_proba(Xte)[:, 1]
print(f"test AUC {roc_auc_score(yte, p):.3f} | accuracy at 0.5 {accuracy_score(yte, p ⩾
.5):.3f}")
print(confusion_matrix(yte, p ⩾ 0.5))
print(classification_report(yte, p ⩾ 0.5, target_names=["repaid", "default"], digits=3))
```

OUTPUT

```
best: {'metric': 'manhattan', 'n_neighbors': 51, 'weights': 'distance'} | CV AUC
0.772
test AUC 0.810 | accuracy at 0.5 0.733
[[134   9]
 [ 47  20]]
              precision    recall  f1-score   support
```

```
              repaid       0.740     0.937     0.827       143
             default       0.690     0.299     0.417        67

            accuracy                           0.733       210
           macro avg       0.715     0.618     0.622       210
        weighted avg       0.724     0.733     0.696       210
```

Python

```
t = 0.3                                                  # lower threshold: missing a
defaulter is costlier
print(f"threshold {t}:"); print(confusion_matrix(yte, p ⩾ t));
print(classification_report(yte, p ⩾ t, target_names=["repaid", "default"], digits=3))
```

OUTPUT

```
threshold 0.3:
[[94 49]
 [ 9 58]]
              precision    recall  f1-score   support

      repaid      0.913     0.657     0.764       143
     default      0.542     0.866     0.667        67

    accuracy                          0.724       210
   macro avg      0.727     0.762     0.715       210
weighted avg      0.794     0.724     0.733       210
```

KNN vs logistic regression

Python

```
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import cross_val_score
cv = StratifiedKFold(5, shuffle=True, random_state=0)
for name, m in [("KNN (tuned)", gs.best_estimator_), ("logistic regression",
make_pipeline(StandardScaler(), LogisticRegression(max_iter=1000)))]:
    s = cross_val_score(m, X, y, cv=cv, scoring="roc_auc"); print(f"{name:20s} 5-fold AUC
{s.mean():.3f} ± {s.std():.3f}")
```

OUTPUT

```
KNN (tuned)          5-fold AUC 0.786 ± 0.046
logistic regression  5-fold AUC 0.803 ± 0.042
```

On this small, roughly linear credit problem, logistic regression is as good or better, and far easier to explain. KNN shines when the boundary is irregular and the data are dense.

Scoring new applicants and explaining with neighbours

Python

```
best = gs.best_estimator_.fit(X, y)
new_apps["p_default"] = best.predict_proba(new_apps[feats])[:, 1].round(3)
print(new_apps.p_default.describe().round(3).to_dict())
app = new_apps.sort_values("p_default", ascending=False).iloc[0]
dist_, idx = best[-1].kneighbors(best[0].transform(app[feats].to_frame().T.astype(float)),
n_neighbors=5)
print(f"{app.customer_id}: P(default) {app.p_default:.2f}. Its 5 most similar past
customers:")
print(known.iloc[idx[0]][feats + ["default"]].to_string(index=False))
```

OUTPUT

```
{'count': 150.0, 'mean': 0.295, 'std': 0.176, 'min': 0.016, '25%': 0.168, '50%':
0.271, '75%': 0.401, 'max': 0.802}
L0782: P(default) 0.80. Its 5 most similar past customers:
 age  ed  employ  address  income  debtinc  creddebt  othdebt  default
  35   2       1        5    33.7     30.7      3.24     7.11        1
  47   2       0        4    26.5     32.8      4.28     4.41        1
  40   1       3        4    49.7     24.1      4.34     7.64        1
  39   1       5        4    32.7     24.3      3.44     4.51        1
  26   1       1        2    52.2     29.4      5.63     9.72        1
```

KNN explanations are concrete: here, all five of the most similar past customers defaulted.

KNN regression

Python

```
from sklearn.neighbors import KNeighborsRegressor
from sklearn.metrics import mean_absolute_error
h = pd.read_csv("house_prices.csv").dropna(subset=["area_sqft"])
hf = ["area_sqft", "bedrooms", "bathrooms", "age_years", "distance_to_center_km"]
Htr, Hte, htr, hte = train_test_split(h[hf], h.price_lakh, test_size=0.2, random_state=0)
for k in (1, 5, 15, 40):
    m = make_pipeline(StandardScaler(), KNeighborsRegressor(k,
weights="distance")).fit(Htr, htr)
    print(f"k = {k:2d}: test MAE {mean_absolute_error(hte, m.predict(Hte)):.2f} lakh")
```

OUTPUT

```
k =  1: test MAE 16.40 lakh
k =  5: test MAE 12.57 lakh
k = 15: test MAE 12.96 lakh
k = 40: test MAE 12.42 lakh
```

Strengths, weaknesses and the curse of dimensionality

Strengths	Weaknesses
Simple, intuitive, no training	Slow predictions on large data (must search all points)
Naturally multiclass; non-linear boundaries	Needs scaling; sensitive to irrelevant features
Explanations by example	Struggles in high dimensions
Works for regression too	Memory: stores the whole training set

Python

```python
rng = np.random.default_rng(0)
for dims in (2, 10, 100, 1000):
    pts_ = rng.uniform(size=(500, dims)); d = np.linalg.norm(pts_[1:] - pts_[0], axis=1)
    print(f"{dims:4d} dimensions: nearest/farthest distance ratio {d.min() / d.max():.2f}")
```

OUTPUT

```
   2 dimensions: nearest/farthest distance ratio 0.01
  10 dimensions: nearest/farthest distance ratio 0.30
 100 dimensions: nearest/farthest distance ratio 0.73
1000 dimensions: nearest/farthest distance ratio 0.90
```

In high dimensions every point is almost equally far from every other (the ratio approaches 1), so "nearest" stops meaning "similar". Reduce dimensions (feature selection, PCA) first. For millions of vectors (embeddings in RAG systems, GenAI track), approximate nearest-neighbour indexes (FAISS, HNSW) make search fast.

Summary

KEY TAKEAWAYS

- KNN predicts from the k most similar labelled examples: majority vote or (weighted) average.
- Distances: Euclidean, Manhattan, Minkowski, cosine; always scale features first.
- Small k overfits, large k underfits; tune k, weights and metric with cross-validation.
- Bank Loans: a scaled, tuned KNN reaches a decent AUC, and logistic regression is as good and more interpretable on this data.
- KNN suffers from slow prediction and the curse of dimensionality; approximate nearest-neighbour search powers vector databases.

Quick quiz

6 questions · tap an answer to check it

1. What does KNN do during training?

A Stores the training data (lazy learning)

B Fits coefficients by gradient descent

C Builds a tree

D Computes clusters

Answer: A. Work happens at prediction time.

2. Why must features be scaled for KNN?

- **A** Features with large ranges would dominate the distance
- **B** To make it faster
- **C** Because KNN needs normal data
- **D** It is optional

Answer: A. Income would swamp education level.

3. What happens with k = 1?

- **A** The model memorises the training data and tends to overfit
- **B** It underfits
- **C** It always predicts the majority class
- **D** It is invalid

Answer: A. Training accuracy is 100%.

4. Which distance is Minkowski with p = 1?

- **A** Euclidean
- **B** Manhattan
- **C** Cosine
- **D** Hamming

Answer: B. Sum of absolute differences.

5. What is the curse of dimensionality for KNN?

- **A** In many dimensions, all points become roughly equally distant, so neighbours are not meaningful
- **B** More rows slow it down
- **C** It needs more classes
- **D** Scaling stops working

Answer: A. Reduce dimensions first.

6. How does `weights="distance"` change predictions?

- **A** Closer neighbours get more influence than farther ones
- **B** All neighbours count equally
- **C** It removes far neighbours
- **D** It changes the metric

Answer: A. Weights are 1/distance.

EXERCISE

Practice

(1) Use KNN to predict churn on `telecom_churn.csv` (one-hot + scale) and tune k with GridSearchCV; compare AUC with logistic regression. (2) Plot test MAE against k (1–50) for KNN regression on house prices. (3) Find the 5 nearest neighbours of a new house (1,400 sq ft, 3 bed, 2 bath, 8 years, 6 km).

Show solution (worked answer)

Python

```
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import OneHotEncoder
tc = pd.read_csv("telecom_churn.csv"); Xc = tc.drop(columns=["customer_id", "churn",
"total_charges"]); yc = (tc.churn == "Yes").astype(int)
pre = ColumnTransformer([("n", StandardScaler(), ["senior_citizen", "tenure_months",
"monthly_charges", "support_calls"]), ("c", OneHotEncoder(), ["contract",
"internet_service", "payment_method"])])
g = GridSearchCV(make_pipeline(pre, KNeighborsClassifier()),
{"kneighborsclassifier__n_neighbors": range(5, 120, 10)}, cv=5,
scoring="roc_auc").fit(Xc, yc); print(g.best_params_, g.best_score_)
maes = [mean_absolute_error(hte, make_pipeline(StandardScaler(),
KNeighborsRegressor(k)).fit(Htr, htr).predict(Hte)) for k in range(1, 51)]
m = make_pipeline(StandardScaler(), KNeighborsRegressor(5)).fit(h[hf], h.price_lakh)
print(m.predict(pd.DataFrame([[1400, 3, 2, 8, 6]], columns=hf)))
```

INTERVIEW CORNER

Questions you may be asked

Q: How does KNN work and how do you choose k? It finds the k closest training points by a distance metric and votes or averages; choose k (and weighting, metric) by cross-validation, balancing overfitting (small k) and oversmoothing (large k).

Q: What are the disadvantages of KNN? Slow and memory-heavy predictions, sensitivity to scaling and irrelevant features, poor performance in high dimensions, and imbalanced classes skew votes.

Q: How would you speed up nearest-neighbour search on millions of points? KD-trees or ball trees for low dimensions, and approximate nearest-neighbour indexes (HNSW, FAISS, ScaNN) for high-dimensional embeddings.

Practise more online

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